EEE 101 (Electrical Circuits I): Complete Definitions, Laws, Theorems, Proofs & Short Questions Solutions

DC Circuits · Network Theorems · First- & Second-Order Transients · Magnetic & AC Circuits

With Full Derivations, Working Formulas & Textbook Citations (Sadiku, Nilsson, Dorf & Boylestad)

Based on BUET L-1/T-1 EEE Examinations (2010–2025)

Comprehensive BUET Exam Preparation Reference

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Part I

Fundamental Concepts, Basic Laws, Sources & Resistive Networks (CO1)

1 Charge, Current, Voltage, Power, Energy & Circuit Sources

1.1 Definitions of Current, Voltage & Power, and Their Interrelation

Exam Appearances: `2011-12 Q5a`, `2013-14 Q1b-iii`, `2014-15 Q1b`, `2019-20 Q1a (15)`, `2020-21 Q2b`, `2021-22 Q7b`, `2024-25 Q5a (20)`.
Textbook Citations: C. K. Alexander & M. N. O. Sadiku, *Fundamentals of Electric Circuits* (5th Ed.), Ch. 1, Sec. 1.3–1.5 (pp. 5–13); J. W. Nilsson & S. A. Riedel, *Electric Circuits* (10th Ed.), Ch. 1, Sec. 1.4–1.6; R. C. Dorf & J. A. Svoboda, *Introduction to Electric Circuits* (9th Ed.), Ch. 1, Sec. 1.2–1.5.

1. Definitions of Electric Charge, Current, Voltage, Power, and Energy (`2024-25 Q5a`)

- **Electric Charge (q):** An intrinsic electrical property of the atomic particles of which matter consists, measured in **coulombs (C)**, where 1 electron carries $e = -1.602 \times 10^{-19}$ C (and 1 C $= 6.24 \times 10^{18}$ electrons).
- **Electric Current (i):** The time rate of change (or flow) of electric charge past a given cross-section, measured in **amperes (A)**, where 1 A $= 1$ C/s:

$$\boxed{i(t) \triangleq \frac{dq}{dt}} \qquad \Longleftrightarrow \qquad q(t) = \int_{t_0}^{t} i(\tau)\, d\tau + q(t_0) \tag{1}$$

- **Voltage / Potential Difference (v_{ab}):** The energy (or work w) required to move a unit positive charge (1 C) from a reference point b to point a through an element, measured in **volts (V)**, where 1 V $= 1$ J/C:

$$\boxed{v_{ab} \triangleq \frac{dw}{dq}} \tag{2}$$

- **Power (p) and Energy (w):** Power is the time rate of expending or absorbing energy, measured in **watts (W)**, where 1 W $= 1$ J/s:

$$\boxed{p(t) \triangleq \frac{dw}{dt}} \qquad \Longleftrightarrow \qquad w(t) = \int_{t_0}^{t} p(\tau)\, d\tau = \int_{t_0}^{t} v(\tau) i(\tau)\, d\tau \tag{3}$$

2. Derivation of the Interrelation Between Voltage, Current, and Power (`2024-25 Q5a`)

Applying the chain rule of differentiation to Equation (3) and substituting Equations (2) and (1):

$$p(t) = \frac{dw}{dt} = \left(\frac{dw}{dq}\right) \cdot \left(\frac{dq}{dt}\right) = v(t) i(t) \implies \boxed{p(t) = v(t) i(t)} \tag{4}$$

- **Passive Sign Convention (PSC):** When current i enters the **positive ($+$)** terminal of an element across which voltage v is defined, $p = +vi > 0$ means power is being **delivered to (absorbed by)** the element, whereas $p = +vi < 0$ (or current leaving the $+$ terminal) means power is being **extracted from (supplied/delivered by)** the element.
- **Conservation of Energy / Tellegen's Theorem (`2011-12 Q5a`, `2013-14 Q1b-iii`):** At any instant of time in any electric circuit, the algebraic sum of powers of all elements is zero: $\boxed{\sum p_k = 0 \implies \sum P_{\text{supplied}} = \sum P_{\text{absorbed}}}$.

3. Solved Exam Problem on Power & Energy (`2024-25 Q5a`)

Problem: *The voltage and current at the terminals of an arbitrary circuit element (assuming passive sign convention) are given for $0 < t < 40$ s (and 0 otherwise) as:*

$$v(t) = t(1 - 0.025t) \text{ V} = (t - 0.025t^2) \text{ V}, \qquad i(t) = (4 - 0.2t) \text{ A}$$

Solution: For $0 < t < 40$ s, the instantaneous power delivered to the element is:

$$p(t) = v(t)i(t) = (t - 0.025t^2)(4 - 0.2t) = 4t - 0.3t^2 + 0.005t^3 \text{ W}$$

To find the stationary points (extrema) of $p(t)$, set $\frac{dp}{dt} = 0$:

$$\frac{dp}{dt} = 4 - 0.6t + 0.015t^2 = 0 \implies 15t^2 - 600t + 4000 = 0 \implies 3t^2 - 120t + 800 = 0$$

$$t = \frac{120 \pm \sqrt{120^2 - 4(3)(800)}}{6} = 20 \pm \frac{\sqrt{4800}}{6} = 20 \pm \frac{20}{\sqrt{3}} \text{ s} \implies t_1 \approx 8.453 \text{ s}, \quad t_2 \approx 31.547 \text{ s}$$

Checking the second derivative $\frac{d^2p}{dt^2} = -0.6 + 0.03t$:

1. **(i) Instant of Maximum Power Delivered to the Element:** At $t_1 = 20 - \frac{20}{\sqrt{3}} \approx \boxed{8.453 \text{ s}}$, $\frac{d^2p}{dt^2} = -0.346 < 0$ (maximum positive power absorbed):

$$p_{\text{max,delivered}} = p(8.453) = 4(8.453) - 0.3(8.453)^2 + 0.005(8.453)^3 = \frac{160}{9\sqrt{3}} \approx \boxed{10.264 \text{ W}}$$

2. **(ii) Instant of Maximum Power Extracted from the Element:** At $t_2 = 20 + \frac{20}{\sqrt{3}} \approx \boxed{31.547 \text{ s}}$, $\frac{d^2p}{dt^2} = +0.346 > 0$ (minimum of $p(t)$, i.e., maximum negative power):

$$p(31.547) = -\frac{160}{9\sqrt{3}} \approx -10.264 \text{ W} \implies \text{Maximum power extracted} = \boxed{10.264 \text{ W}}$$

3. **(iii) Net Energy Delivered to the Element at $t = 30$ s:**

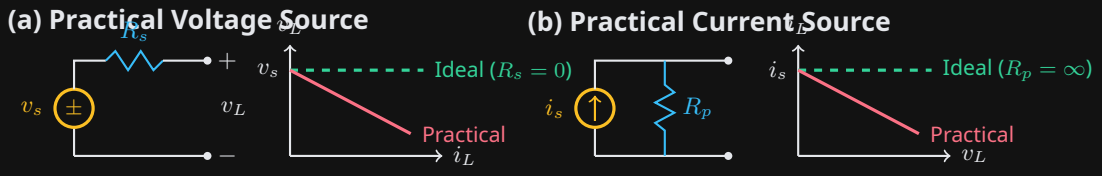
$$w(30) = \int_0^{30} \left(4t - 0.3t^2 + 0.005t^3\right) dt = \left[2t^2 - 0.1t^3 + 0.00125t^4\right]_0^{30} = 1800 - 2700 + 1012.5 = \boxed{112.5 \text{ J}}$$

1.2 Ideal vs. Practical (Actual) Voltage & Current Sources

Exam Appearances: `2018-19 Q2c (5)`, `2019-20 Q1b (15)`.
Textbook Citations: Alexander & Sadiku, Ch. 1, Sec. 1.6 (p. 15) & Ch. 4, Sec. 4.10.1 (pp. 155–157); Nilsson & Riedel, Ch. 1, Sec. 1.3 & Ch. 4, Sec. 4.9; Boylestad, *Introductory Circuit Analysis*, Ch. 8, Sec. 8.2.

1. Distinguish Between Ideal and Practical (Actual) Voltage and Current Sources (2018–19 Q2c)

Feature	Ideal Source	Practical (Actual) Source
1. **Voltage Source**	Maintains a specified terminal voltage $v_L = v_s$ **completely independent** of the current i_L drawn from it. Has **zero internal resistance** ($R_s = 0$).	Modeled as an ideal voltage source v_s **in series** with an internal resistance $R_s > 0$. Terminal voltage drops with load current: $v_L = v_s - i_L R_s = \frac{R_L}{R_s+R_L} v_s$.
2. **Current Source**	Delivers a specified terminal current $i_L = i_s$ **completely independent** of the terminal voltage v_L across it. Has **infinite internal parallel resistance** ($R_p = \infty$).	Modeled as an ideal current source i_s **in parallel** with a finite internal resistance $R_p < \infty$. Load current drops as v_L increases: $i_L = i_s - \frac{v_L}{R_p} = \frac{R_p}{R_p+R_L} i_s$.
3. v-i **Characteristic**	Perfectly flat horizontal line ($v_L = v_s$) or vertical line ($i_L = i_s$); can theoretically supply infinite power.	Sloping straight line with voltage drop $i_L R_s$ or current drop v_L/R_p; maximum power is finite ($P_{\max} = \frac{v_s^2}{4R_s}$).



2. Practical Source Modeling Problem (2019–20 Q1b)

Problem: *A battery provides* 12.5 *V to a radio* ($R_{L1} = 6.25\ \Omega$) *and* 11.7 *V to a light bulb* ($R_{L2} = 0.65\ \Omega$). *Find the Thevenin and Norton equivalents of the battery.*
Solution: By voltage division for a practical battery with Thevenin voltage V_{Th} and internal resistance R_{Th}, $V_L = \frac{R_L}{R_L+R_{\text{Th}}} V_{\text{Th}}$ (or $V_{\text{Th}} = V_L + I_L R_{\text{Th}}$):

$$\text{With radio } (I_{L1} = \frac{12.5}{6.25} = 2\text{ A}): \quad V_{\text{Th}} = 12.5 + 2R_{\text{Th}}$$
$$\text{With bulb } (I_{L2} = \frac{11.7}{0.65} = 18\text{ A}): \quad V_{\text{Th}} = 11.7 + 18R_{\text{Th}}$$

Subtracting the two equations gives $16R_{\text{Th}} = 0.8 \implies \boxed{R_{\text{Th}} = R_N = 0.05\ \Omega}$, and substituting back gives $\boxed{V_{\text{Th}} = 12.6\text{ V}}$ and Norton current $\boxed{I_N = \frac{V_{\text{Th}}}{R_{\text{Th}}} = \frac{12.6}{0.05} = 252\text{ A}}$.

1.3 Ohm's Law, Static vs. Dynamic Resistance & Meter Loading

Exam Appearances: 2015–16 Q1a,b, 2018–19 Q1b, Q3c (5), 2023–24 Q5a,b,c (35).
Textbook Citations: R. L. Boylestad, *Introductory Circuit Analysis* (12th Ed.), Ch. 3; R. L. Boylestad & L. Nashelsky, *Electronic Devices and Circuit Theory* (11th Ed.), Ch. 1, Sec. 1.7–1.8 (DC/Static vs. AC/Dynamic Resistance); Alexander & Sadiku, Ch. 2, Sec. 2.2 & Prob. 2.73 (p. 77).

1. Define Static and Dynamic Resistances with Suitable Examples (2018–19 Q3c)

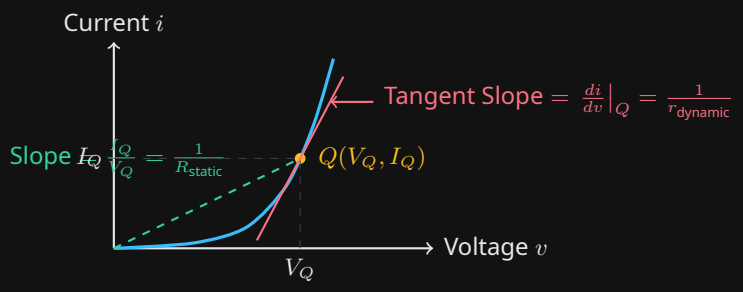
1. **Static (DC or Chord) Resistance (R_{DC} or R_S):** The ratio of the total steady-state (DC) voltage V_Q across a circuit element to the steady-state (DC) current I_Q flowing through it at a fixed operating point (quiescent point $Q(V_Q, I_Q)$) on its i-v characteristic curve:

$$\boxed{R_{\text{static}} = R_{\text{DC}} = \frac{V_Q}{I_Q}} \tag{5}$$

   Geometrically, $1/R_{\text{static}}$ is the slope of the secant (chord) line drawn from the origin $(0, 0)$ to the operating point Q.
2. **Dynamic (AC, Differential, or Incremental) Resistance (r_d):** The rate of change of terminal voltage v with respect to current i for small-signal AC fluctuations around a given operating point Q on the i-v characteristic curve:

$$\boxed{r_{\text{dynamic}} = r_d = \left.\frac{\Delta v}{\Delta i}\right|_Q = \left.\frac{dv}{di}\right|_Q} \tag{6}$$

   Geometrically, $1/r_d$ is the slope of the **tangent line** to the i-v curve at the operating point Q.
3. **Suitable Examples:**
   - *Linear Ohmic Resistor (e.g., carbon-film or metal-film resistor):* Obeys Ohm's law $v = iR$, so its i-v graph is a straight line through the origin and $R_{\text{static}} = r_{\text{dynamic}} = R =$ constant.
   - *Semiconductor PN-Junction Diode:* Obeys Shockley's nonlinear equation $I_D = I_S(e^{V_D/nV_T} - 1)$. At $V_D = 0.7$ V and $I_D = 10$ mA, its static resistance is $R_{\text{DC}} = \frac{0.7\text{ V}}{10\text{ mA}} = 70\ \Omega$, whereas its dynamic resistance is $r_d = \frac{nV_T}{I_D} \approx \frac{26\text{ mV}}{10\text{ mA}} = 2.6\ \Omega$.
   - *Incandescent Tungsten Lamp Filament / Tunnel Diode:* A tungsten filament's resistance increases nonlinearly with current/temperature, while a tunnel diode exhibits negative dynamic resistance ($r_d = \frac{dv}{di} < 0$) in its valley region even though its static resistance V/I is always positive.



2. Ammeter Half-Deflection Problem (2023–24 Q5b — Sadiku Ch. 2, Prob. 2.73, p. 77)

Problem: *An ammeter model consists of an ideal ammeter in series with a $20\ \Omega$ resistor. It is connected in parallel with a current source I and an unknown resistor R_x. When a potentiometer R is added in series with the ammeter branch and adjusted until the ammeter reading drops to one-half its previous reading, $R = 65\ \Omega$. What is the value of R_x?*
Solution:

1. **Before adding R:** The ammeter branch (resistance $R_m = 20\ \Omega$) is in parallel with R_x. By current division, the initial ammeter reading is:

$$I_{A1} = \frac{R_x}{R_x + 20} I$$

2. **After adding $R = 65\ \Omega$ in series with the ammeter:** The total resistance of the ammeter branch becomes $20 + R = 20 + 65 = 85\ \Omega$. By current division, the new ammeter reading is:

$$I_{A2} = \frac{R_x}{R_x + 85} I$$

3. Since $I_{A2} = \frac{1}{2} I_{A1}$:

$$\frac{R_x}{R_x + 85} I = \frac{1}{2} \left(\frac{R_x}{R_x + 20} I \right) \implies 2(R_x + 20) = R_x + 85 \implies \boxed{R_x = 45\ \Omega}$$

2 Network Topology, Kirchhoff's Laws, Dividers & Y-△ Transformations

2.1 Kirchhoff's Laws (KCL & KVL) and Voltage Division Applicability

Exam Appearances: `2010-11 Q2a (5)`, `2021-22 Q5b`, `2022-23 Q5a`.
Textbook Citations: Alexander & Sadiku, Ch. 2, Sec. 2.3–2.6 (pp. 35–47); Nilsson & Riedel, Ch. 2, Sec. 2.4 & Ch. 3, Sec. 3.3.

1. Fundamental Definitions of Network Topology

- **Branch (b):** A single two-terminal circuit element (such as a voltage source, current source, resistor, inductor, or capacitor).
- **Node (n):** The point of connection between two or more branches.
- **Loop (l) and Mesh:** Any closed path in a circuit starting at a node, passing through a set of nodes, and returning to the starting node without passing through any node more than once. A **mesh** is a loop that does not contain any other loop within it. For a connected network with b branches, n nodes, and l independent loops, the **Fundamental Theorem of Network Topology** states:

$$\boxed{b = l + n - 1} \tag{7}$$

2. Statements and Proofs of Kirchhoff's Current Law (KCL) and Voltage Law (KVL) (`2021-22 Q5b`, `2022-23 Q5a`)

1. **Kirchhoff's Current Law (KCL):** *"The algebraic sum of all currents entering a node (or a closed boundary) at any instant is zero"* (or equivalently, the sum of currents entering a node equals the sum of currents leaving the node):

$$\boxed{\sum_{k=1}^{N} i_k(t) = 0 \quad \Longleftrightarrow \quad \sum i_{\text{entering}} = \sum i_{\text{leaving}}} \tag{8}$$

   Physical Basis & Proof: KCL is based on the **law of conservation of electric charge**. If currents $i_1(t), i_2(t), \ldots, i_N(t)$ enter a node, the net current entering is $i_T(t) = \sum_{k=1}^{N} i_k(t)$. Integrating with respect to time gives the net charge accumulated at the node, $q_T(t) = \sum_{k=1}^{N} q_k(t)$. Because a mathematical node cannot store net electric charge ($q_T(t) = 0$), differentiating gives $i_T(t) = \sum_{k=1}^{N} i_k(t) = 0$.

2. **Kirchhoff's Voltage Law (KVL):** *"The algebraic sum of all branch voltages around any closed path (or loop) at any instant is zero"* (or equivalently, the sum of voltage drops around a loop equals the sum of voltage rises):

$$\boxed{\sum_{m=1}^{M} v_m(t) = 0 \quad \Longleftrightarrow \quad \sum v_{\text{drops}} = \sum v_{\text{rises}}} \tag{9}$$

   Physical Basis & Proof: KVL is based on the **law of conservation of energy** and the conservative nature of electrostatic potential. Moving a test charge q around a closed loop back to its starting point requires zero net work ($\oint dw = 0$), so dividing by q gives $\sum v_m = \sum \frac{\Delta w_m}{q} = 0$.

3. Why Voltage Division Cannot Be Used to Find v_1 in `2010-11 Q2a`

In `2010-11 Q2a` (a BJT bias circuit where a 10 V source is connected to two 20 kΩ resistors in a voltage-divider configuration, with the middle node connected to a 0.7 V base-emitter source leading to the rest of the circuit), **the simple voltage-divider formula $v_1 = \frac{R_2}{R_1+R_2}(10\text{ V})$ cannot be used** because:

- The voltage-divider rule is derived under the strict assumption that the two resistors R_1 and R_2 are connected **exclusively in series** so that they carry the **exact same current** ($i_{R1} = i_{R2}$).
- In Fig. Q. 2(a), a third branch containing the 0.7 V source is connected to the junction node between the two 20 kΩ resistors and draws a non-zero branch current i_b away from the node. By KCL, the currents through the two 20 kΩ resistors are unequal ($i_{R1} = i_{R2} + i_b \neq i_{R2}$), invalidating the series voltage-divider formula (a classic loading effect).

2.2 Wye-Delta (Y-Δ) and Delta-Wye (Δ-Y) Transformations

Exam Appearances: `2010-11 Q3a`, `2015-16 Q1c (11)`, `2018-19 Q3a (15)`, `2024-25 Q8c (5)`.
Textbook Citations: Alexander & Sadiku, Ch. 2, Sec. 2.7 (pp. 52–57) & Ch. 12; Nilsson & Riedel, Ch. 3, Sec. 3.7.

1. Conversion Formulas and Derivation of Wye-to-Delta (Y-Δ) Transformation (`2018-19 Q3a`)

Consider a three-terminal Wye (Y or T) network with resistors R_1, R_2, R_3 connected to terminals a, b, c and neutral node n, and an equivalent Delta (Δ or Π) network with resistors R_a, R_b, R_c opposite to R_1, R_2, R_3 respectively. Equating the open-circuit resistances between each pair of terminals (ab, bc, ca) in the Y and Δ networks:

$$R_{ab}(Y) = R_{ab}(\Delta) \implies R_1 + R_2 = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} \tag{10}$$

$$R_{bc}(Y) = R_{bc}(\Delta) \implies R_2 + R_3 = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c} \tag{11}$$

$$R_{ca}(Y) = R_{ca}(\Delta) \implies R_3 + R_1 = \frac{R_b(R_c + R_a)}{R_a + R_b + R_c} \tag{12}$$

Subtracting Equation (11) from Equation (10) gives $R_1 - R_3 = \frac{R_b(R_c - R_a)}{R_a+R_b+R_c}$. Adding this to Equation (12) and dividing by 2 yields the **Delta-to-Wye ($\Delta \to$ Y) conversion formulas**:

$$\boxed{R_1 = \frac{R_bR_c}{R_a + R_b + R_c}, \qquad R_2 = \frac{R_cR_a}{R_a + R_b + R_c}, \qquad R_3 = \frac{R_aR_b}{R_a + R_b + R_c}} \tag{13}$$

Rule: Each resistor in the Y network is the product of the resistors in the two adjacent Δ branches divided by the sum of the three Δ resistors.

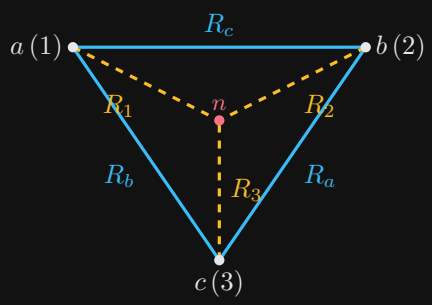
To convert a general Y load into its equivalent Δ (`2018-19 Q3a`), multiply the equations in (13) in pairs and sum them:

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_aR_bR_c(R_a + R_b + R_c)}{(R_a + R_b + R_c)^2} = \frac{R_aR_bR_c}{R_a + R_b + R_c}$$

Dividing this identity by R_1, R_2, and R_3 in Equation (13) gives the **Wye-to-Delta (Y $\to \Delta$) conversion formulas**:

$$\boxed{R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}, \qquad R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}, \qquad R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}} \tag{14}$$

Rule: Each resistor in the Δ network is the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistor. For a **balanced network**, $\boxed{R_\Delta = 3R_Y \iff R_Y = \frac{1}{3}R_\Delta}$.



2. Practical Uses of Wye-Delta (Y-Δ) Transformation (`2015-16 Q1c`, `2024-25 Q8c`)

1. **Simplifying Bridge and Non-Series-Parallel Networks:** Reduces unbalanced Wheatstone bridges and complex planar/non-planar resistive/impedance networks (where elements are neither in series nor in parallel) into simple series-parallel combinations to compute equivalent resistance R_{eq}.
2. **Three-Phase AC Power System Analysis:** Converts three-phase Δ-connected loads and generators into equivalent Y-connected systems ($Z_Y = Z_\Delta/3$) so that balanced three-phase power systems can be analyzed using a single-phase equivalent circuit.
3. **Three-Phase Induction Motor Starting (Y-Δ Starters):** Used in industrial motor starters to start a three-phase induction motor in Wye (Y) configuration (reducing starting phase voltage by $1/\sqrt{3}$ and starting line current to $1/3$ of full-load delta current) before switching to Delta (Δ) for normal full-power operation.
4. **Passive Filter, Attenuator, and Matching Network Design:** Converts between T-section (Y) and Π-section (Δ) attenuators, passive filters, and two-port impedance-matching networks.

2.3 Classic Analytical & Topological Short Problems

Exam Appearances: `2017-18 Q5a (11)`, `2017-18 Q5b (6)`, `2017-18 Q8a (7)`, `2018-19 Q1c (15)`.

1. Sliding Load Between Two Voltage Sources Proof (`2018-19 Q1c`)

Problem: *Two voltage sources v_1 and v_2 are connected by two parallel conductors of length L, each having resistance $r\ \Omega/m$. A load resistor R is placed at distance x from v_1 (and $L-x$ from v_2). (i) Show that the load voltage is $v = \frac{v_1 RL + R(v_2 - v_1)x}{RL + 2rLx - 2rx^2}$, and (ii) find x for which v is minimum.*
Proof:

1. **Part (i):** The top and bottom conductor segments to the left of R (length x) have total series resistance $R_{\text{left}} = rx + rx = 2rx$. Similarly, the top and bottom conductor segments to the right of R (length $L-x$) have total series resistance $R_{\text{right}} = 2r(L-x)$. Applying KCL at the top node of load resistor R (voltage v):

$$\frac{v - v_1}{2rx} + \frac{v}{R} + \frac{v - v_2}{2r(L-x)} = 0$$

Multiplying through by $2rRx(L-x)$:

$$R(L-x)(v-v_1)+2rx(L-x)v+Rx(v-v_2)=0$$
$$[R(L-x)+Rx+2rx(L-x)]\,v=v_1R(L-x)+v_2Rx$$
$$(RL+2rLx-2rx^2)v=v_1RL+R(v_2-v_1)x \implies \boxed{v=\frac{v_1RL+R(v_2-v_1)x}{RL+2rLx-2rx^2}}$$

2. **Part (ii):** Differentiating $v(x)=\frac{N(x)}{D(x)}$ with respect to x and setting $\frac{dv}{dx}=0$ (i.e., $D(x)N'(x)-N(x)D'(x)=0$):

$$(RL+2rLx-2rx^2)\,[R(v_2-v_1)]-[v_1RL+R(v_2-v_1)x]\,(2rL-4rx)=0$$
$$2r(v_2-v_1)x^2+4rv_1Lx+\left[RL(v_2-v_1)-2rv_1L^2\right]=0$$
$$(v_2-v_1)x^2+2v_1Lx+L^2\left[\frac{R}{2rL}(v_2-v_1)-v_1\right]=0$$

Applying the quadratic formula for x:

$$x=\frac{-2v_1L\pm\sqrt{4v_1^2L^2-4(v_2-v_1)L^2\left[\frac{R}{2rL}(v_2-v_1)-v_1\right]}}{2(v_2-v_1)}$$
$$=\boxed{\frac{L}{v_2-v_1}\left[-v_1\pm\sqrt{v_1v_2-\frac{R}{2rL}(v_1-v_2)^2}\right]}$$

2. Infinite Ladder Network Input Resistance (2017-18 Q5a)

Problem: *Find the equivalent input resistance R_{in} of the infinite ladder network where each section has a series resistor R on the top rail, a shunt resistor R in the middle, and a series resistor R on the bottom rail, with $R=1706\ \Omega$.*
Solution: Because the ladder is infinite, adding one more identical section (R top, R shunt, R bottom) in front of the existing infinite network leaves the input resistance R_{in} unchanged:

$$R_{\text{in}}=R+R+(R\parallel R_{\text{in}})=2R+\frac{RR_{\text{in}}}{R+R_{\text{in}}}$$
$$R_{\text{in}}(R+R_{\text{in}})=2R(R+R_{\text{in}})+RR_{\text{in}} \implies R_{\text{in}}^2-2RR_{\text{in}}-2R^2=0$$

Solving the quadratic equation for $R_{\text{in}}>0$:

$$\boxed{R_{\text{in}}=(1+\sqrt{3})R\approx 2.73205\,R} \tag{15}$$

For $R=1706\ \Omega$, $\boxed{R_{\text{in}}=1706(1+\sqrt{3})\approx 4660.88\ \Omega}$. *(Note: If each ladder section has only a single series resistor R and a single shunt resistor R, then $R_{\text{in}}=R+(R\parallel R_{\text{in}}) \implies R_{\text{in}}=\frac{1+\sqrt{5}}{2}R\approx 2760.37\ \Omega$).*

3. Three-Terminal Black Box with Five $1\ \Omega$ Resistors (2017-18 Q8a)

Problem: *A three-terminal black box (A,B,C) contains five $1\ \Omega$ resistors. Ohmmeter measurements give $R_{AB}=1.5\ \Omega$, $R_{BC}=3\ \Omega$, and $R_{AC}=2.5\ \Omega$. Find the configuration of the five resistors.*
Solution: Represent the three-terminal box as an equivalent Wye (Y) network with arms R_A,R_B,R_C joined at an internal node N:

$$R_{AB}=R_A+R_B=1.5\ \Omega$$
$$R_{BC}=R_B+R_C=3.0\ \Omega$$
$$R_{AC}=R_A+R_C=2.5\ \Omega$$

Adding all three equations gives $2(R_A + R_B + R_C) = 7.0\ \Omega \implies R_A + R_B + R_C = 3.5\ \Omega$. Subtracting each pair yields:

$$\boxed{R_A = 0.5\ \Omega = (1\ \Omega \parallel 1\ \Omega), \qquad R_B = 1.0\ \Omega = (\text{one } 1\ \Omega \text{ resistor}), \qquad R_C = 2.0\ \Omega = (1\ \Omega + 1\ \Omega \text{ in series})}$$

Total number of $1\ \Omega$ resistors used $= 2 + 1 + 2 = 5$ resistors!

4. Cut-Set KCL & KVL Short Problem (`2017-18 Q5b`)

Problem: *In Fig. for Q. 5(b), the left subnetwork and right subnetwork are linked by a top $2\ \Omega$ resistor (between V_A and V_B) and a bottom parallel combination of a 2 A current source (pointing from C to D) and a $1\ \Omega$ resistor (between V_C and V_D). If $V_A - V_B = 6$ V, find $V_C - V_D$.*
Solution:

1. The current flowing from left to right through the top $2\ \Omega$ bridge resistor is $I_{AB} = \frac{V_A - V_B}{2\ \Omega} = \frac{6\ \text{V}}{2\ \Omega} = 3$ A.
2. Enclose the entire right subnetwork in a closed boundary (cut-set). By KCL for a closed boundary, the total current entering the right subnetwork through the top wire ($I_{AB} = 3$ A) must equal the total current returning to the left subnetwork through the bottom branch (CD):

$$I_{AB} + I_{CD} = 0 \implies 3\ \text{A} + \left(2\ \text{A} + \frac{V_C - V_D}{1\ \Omega}\right) = 0 \implies \boxed{V_C - V_D = -5\ \text{V}}$$

Part II

Methods of Analysis & Network Theorems (CO1 & CO2)

3 Supernode, Supermesh, Linearity & Superposition Theorem

3.1 Supernode, Supermesh & Reconstructing a Circuit from Mesh Equations

Exam Appearances: `2015-16 Q2a (15)`, `2015-16 Q4c (5)`.
Textbook Citations: Alexander & Sadiku, Ch. 3, Sec. 3.3 (p. 89) & Sec. 3.5 (p. 98).

1. Define Supernode and Supermesh (`2015-16 Q4c`)

- **Supernode:** A **supernode** is formed by enclosing a (dependent or independent) voltage source connected between two non-reference nodes and any elements connected in parallel with it inside a closed boundary.
  - *Properties:* A supernode has no single voltage of its own; it requires applying **both KCL** (to the closed boundary enclosing the voltage source) and **KVL** (to the voltage source branch to provide the constraint equation relating the two node voltages).
- **Supermesh:** A **supermesh** results when two meshes have a (dependent or independent) current source in common. The supermesh is created by excluding the shared current source and any elements connected in series with it, forming a larger loop around the periphery of the two meshes.
  - *Properties:* A supermesh has no single mesh current of its own; it requires applying **both KVL** around the outer periphery of the supermesh and **KCL** at a node on the shared branch to relate the two mesh currents to the current source.

2. Reconstructing a Circuit from Mesh Equations (`2015-16 Q2a`)

Problem: *In solving for currents using mesh analysis, the following equations are obtained. Draw the circuit and find the currents:*

$$i_2 = -10 \text{ A}, \qquad 10i_1 - 22i_2 + 10i_3 = 0, \qquad i_3 = 12 \text{ A}$$

Solution:

1. **Finding the Currents:** Substituting $i_2 = -10$ A and $i_3 = 12$ A into the second equation gives:

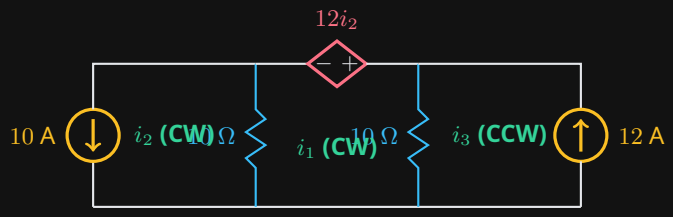
   $$10i_1 - 22(-10) + 10(12) = 0 \implies 10i_1 + 220 + 120 = 0 \implies \boxed{i_1 = -34 \text{ A}, \quad i_2 = -10 \text{ A}, \quad i_3 = 12 \text{ A}}$$

2. **Reconstructing the Circuit:** Rewrite the middle KVL equation (for Mesh 1 or a Supermesh) in standard KVL form:

   $$10(i_1 - i_2) + 10(i_3 - i_2) - 2i_2 = 0$$

   Or, viewing $10i_1 - 22i_2 + 10i_3 = 0$ as the standard inspection KVL equation $-R_{21}i_1 + R_{22}i_2 - R_{23}i_3 = 0$ (multiplied by -1) for a central Mesh 2 sharing a $10\,\Omega$ resistor with Mesh 1 ($R_{12} = 10\,\Omega$), a $10\,\Omega$ resistor with Mesh 3 ($R_{23} = 10\,\Omega$), and having an unshared $2\,\Omega$ resistor ($R_{22} = 10 + 10 + 2 = 22\,\Omega$), while outer Meshes 1 and 3 contain independent current sources forcing i_2 or i_3! Even simpler: let Mesh 2 have an independent 10 A current source in its outer branch

pointing counterclockwise ($i_2 = -10$ A), Mesh 3 have a 12 A current source in its outer branch pointing clockwise ($i_3 = 12$ A), and let the equation $10(i_1 - i_2) - 12i_2 + 10(i_3) = 0$ (or a 3-mesh circuit with a supermesh) represent KVL. The cleanest textbook reconstruction is shown below:



(Check: In the circuit above, $i_2 = -10$ A, $i_3 = 12$ A, and KVL around the middle mesh i_1 gives $10(i_1 - i_2) - 12i_2 + 10(i_1 + i_3 - i_1) = 10i_1 - 22i_2 + 10i_3 = 0$!)

3.2 Linear Circuit, Linearity Property & Superposition Theorem

Exam Appearances: `2015-16 Q4b (15)`, `2018-19 Q3b (15)`, `2021-22 Q7a (22)`, `2022-23 Q5b (18)`.
Textbook Citations: Alexander & Sadiku, Ch. 4, Sec. 4.2–4.3 (pp. 128–131); Nilsson & Riedel, Ch. 4, Sec. 4.13.

1. What Is a Linear Circuit? Explain the Linearity Property (`2018-19 Q3b`, `2021-22 Q7a`, `2022-23 Q5b`)

- **Linearity Property:** Linearity is the property of an element describing a linear relationship between cause (input/excitation) and effect (output/response). It is a combination of two mathematical properties:
  1. **Homogeneity (Scaling) Property:** If the input (excitation) i is multiplied by a constant k, then the output (response) v is multiplied by the same constant k: $f(ki) = kf(i)$ (e.g., $kiR = kv$).
  2. **Additivity Property:** The response to a sum of inputs is the sum of the responses to each input applied separately: $f(i_1+i_2) = f(i_1)+f(i_2)$ (e.g., $(i_1+i_2)R = i_1R+i_2R = v_1+v_2$).
- **Definition of a Linear Circuit:** A **linear circuit** is a circuit whose output is linearly related (or directly proportional) to its input. It consists exclusively of **linear passive elements** (linear resistors, inductors, capacitors), **linear dependent sources**, and **independent sources**.

2. Statement, Steps, Usability & Limitations of the Superposition Principle (`2015-16 Q4b`, `2021-22 Q7a`, `2022-23 Q5b`)

- **Statement of the Superposition Principle:** *"The voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone, with all other independent sources turned off."*
- **Procedure / Rules:**
  1. Consider one independent source at a time while turning off (killing/deactivating) all other independent sources: replace every other **independent voltage source by 0 V (a short circuit)** and every other **independent current source by 0 A (an open circuit)**.
  2. **Dependent sources must be left intact** because they are controlled by circuit variables.

3. Sum the individual voltage or current responses algebraically: $v = \sum_{k=1}^{N} v_k$ and $i = \sum_{k=1}^{N} i_k$.

- **Usability & Limitations (`2015-16 Q4b`):**
  - *Usability:* Breaks a complex multi-source circuit into simpler single-source circuits; essential for analyzing AC circuits driven simultaneously by sources operating at **different frequencies** (e.g., DC + AC, or $\omega_1 \neq \omega_2$ as in `2016-17 Q3b`, `2017-18 Q3a`, `2018-19 Q5a`, and Fourier series analysis), where a single phasor domain cannot be used for the whole circuit at once.
  - *Limitation on Power:* Superposition is based on linearity and **cannot be applied directly to power calculations** because power is a quadratic (nonlinear) function of voltage or current:

$$p = R(i_1 + i_2)^2 = Ri_1^2 + Ri_2^2 + 2Ri_1i_2 = p_1 + p_2 + 2Ri_1i_2 \neq p_1 + p_2$$

    To find power using superposition, one must first find the total current $i = i_1 + i_2$ (or total voltage $v = v_1 + v_2$) by superposition and then compute $p = i^2R = v^2/R$.

4 Thevenin's, Norton's, Maximum Power, Reciprocity & Millman's Theorems

4.1 Thevenin's & Norton's Theorems: Statements, Proofs & Black-Box Problems

Exam Appearances: 2016-17 Q7a, Q8a, 2017-18 Q6c (8), 2021-22 Q8a (8), 2022-23 Q8a (10).
Textbook Citations: Alexander & Sadiku, Ch. 4, Sec. 4.5–4.7 (pp. 139–150).

1. Statements and Proofs of Thevenin's and Norton's Theorems (2021-22 Q8a, 2022-23 Q8a)

- **Statement of Thevenin's Theorem:** *"A linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} (or impedance Z_{Th}), where $V_{Th} = v_{oc}$ is the open-circuit voltage at the terminals and $R_{Th} = R_{in}$ is the input or equivalent resistance at the terminals when all independent sources are turned off."*
- **Statement of Norton's Theorem:** *"A linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N (or impedance Z_N), where $I_N = i_{sc}$ is the short-circuit current through the terminals and $R_N = R_{Th}$ is the input or equivalent resistance at the terminals when all independent sources are turned off ($I_N = V_{Th}/R_{Th}$)."*

Proof of Thevenin's Theorem (via Superposition, Sadiku Sec. 4.7, p. 149): Consider a linear two-terminal circuit containing resistors, dependent sources, and internal independent sources ($v_{s1}, v_{s2}, \ldots, i_{s1}, i_{s2}, \ldots$). Suppose an external current source i is connected across terminals a-b (injecting current i into terminal a). By the principle of superposition, the terminal voltage v across a-b is a linear combination of the external current i and all internal independent sources:

$$v = A_0 i + \underbrace{(A_1 v_{s1} + A_2 v_{s2} + A_3 i_{s1} + \ldots)}_{B_0} = A_0 i + B_0$$

1. **Open-Circuit Condition ($i = 0$):** When terminals a-b are open-circuited, $i = 0$, so $v = v_{\text{oc}} = B_0$. By definition, $v_{\text{oc}} = V_{\text{Th}}$, so $B_0 = V_{\text{Th}}$.
2. **Dead-Network Condition (Internal Independent Sources Turned Off):** When all internal independent sources are set to zero, $B_0 = 0$, and the circuit reduces to its equivalent input resistance R_{Th}, so $v = R_{\text{Th}} i = A_0 i \implies A_0 = R_{\text{Th}}$.

Substituting $A_0 = R_{\text{Th}}$ and $B_0 = V_{\text{Th}}$ gives the exact terminal v-i relationship of the Thevenin equivalent circuit:

$$\boxed{v = R_{\text{Th}} i + V_{\text{Th}}} \qquad (\text{or } v = V_{\text{Th}} - R_{\text{Th}} I_L \text{ for load current } I_L = -i \text{ leaving terminal } a) \qquad (16)$$

2. Coupled Two-Network (A-B) Thevenin Identification (2017-18 Q6c)

Problem: *Two linear resistive networks A and B are connected together. In Connection 1 ($+$ of A to $+$ of B, current i from A to B), $v = 1$ V and $i = 0$ mA. When network B is reversed (Connection 2: $+$ of A to $-$ of B), the new values are $v = 0.5$ V and $i = 0.5$ mA. Find the Thevenin equivalents of A and B.*
Solution: Let network A have Thevenin equivalent (V_A, R_A) and network B have (V_B, R_B):

1. **Connection 1 ($i = 0$ mA, $v = 1$ V):** Since $i = 0$ mA, neither network has any internal voltage drop across R_A or R_B ($v = V_A - iR_A = V_A$ and $v = V_B + iR_B = V_B$). Therefore:

$$\boxed{V_A = 1\,\text{V}} \qquad \text{and} \qquad \boxed{V_B = 1\,\text{V}}$$

2. **Connection 2 (B reversed, $v = 0.5$ V, $i = 0.5$ mA):** Now $v = V_A - iR_A$ and $v = -V_B + iR_B$:

$$\text{For Network } A: \quad 0.5\text{ V} = 1\text{ V} - (0.5\text{ mA})R_A \implies \boxed{R_A = \frac{0.5\text{ V}}{0.5\text{ mA}} = 1\text{ k}\Omega}$$

$$\text{For Network } B: \quad 0.5\text{ V} = -1\text{ V} + (0.5\text{ mA})R_B \implies \boxed{R_B = \frac{1.5\text{ V}}{0.5\text{ mA}} = 3\text{ k}\Omega}$$

4.2 Maximum Power Transfer Theorem (DC, AC & Internal Resistor Optimization)

Exam Appearances: `2013-14 Q4b (17)`, `2015-16 Q2c (5)`, `2016-17 Q4a (15)`, `2017-18 Q3b (18)`, `Q8b (14)`, `2018-19 Q2a (15)`, `2019-20 Q6b (15)`, `2021-22 Q8a (8)`, `2024-25 Q7a (15)`.
Textbook Citations: Alexander & Sadiku, Ch. 4, Sec. 4.8 (pp. 150–152) & Ch. 11, Sec. 11.3 (pp. 464–466); Prob. 4.68 (p. 170).

1. Proof of the DC Maximum Power Transfer Theorem & Applications (`2015-16 Q2c`, `2021-22 Q8a`)

Theorem: *Maximum power is transferred from a linear two-terminal DC network to a variable load resistor R_L when the load resistance equals the Thevenin resistance as seen from the load terminals ($\boxed{R_L = R_{Th}}$).*
Proof: Replace the linear network by its Thevenin equivalent $(V_{\text{Th}}, R_{\text{Th}})$ connected to load R_L. The current through the load is $i = \frac{V_{\text{Th}}}{R_{\text{Th}}+R_L}$, so the power delivered to R_L is:

$$p = i^2 R_L = \left(\frac{V_{\text{Th}}}{R_{\text{Th}} + R_L}\right)^2 R_L = \frac{V_{\text{Th}}^2 R_L}{(R_{\text{Th}} + R_L)^2} \tag{17}$$

To maximize p with respect to the variable load R_L, differentiate Equation (17) with respect to R_L and set $\frac{dp}{dR_L} = 0$:

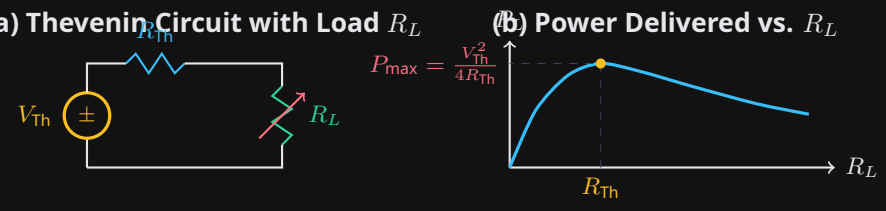
$$\frac{dp}{dR_L} = V_{\text{Th}}^2 \left[\frac{(R_{\text{Th}} + R_L)^2(1) - R_L \cdot 2(R_{\text{Th}} + R_L)}{(R_{\text{Th}} + R_L)^4}\right] = V_{\text{Th}}^2 \left[\frac{R_{\text{Th}} - R_L}{(R_{\text{Th}} + R_L)^3}\right] = 0$$

Thus $R_{\text{Th}} - R_L = 0 \implies \boxed{R_L = R_{\text{Th}}}$ (and $\left.\frac{d^2p}{dR_L^2}\right|_{R_L=R_{\text{Th}}} = -\frac{V_{\text{Th}}^2}{8R_{\text{Th}}^3} < 0$, confirming a maximum). Substituting $R_L = R_{\text{Th}}$ into Equation (17) gives the **maximum power transferred**:

$$\boxed{P_{\text{max}} = \frac{V_{\text{Th}}^2}{4R_{\text{Th}}} = \frac{V_{\text{Th}}^2}{4R_L}} \tag{18}$$

Where We Use the Maximum Power Condition (`2015-16 Q2c`):

- Used extensively in **communications, instrumentation, audio electronics, and signal-processing systems** (e.g., matching a radio/TV receiving antenna to a receiver front-end, matching an audio power amplifier to a loudspeaker, or matching a microphone/sensor transducer to a preamplifier), where the received signal energy is very small and maximizing load power is paramount.
- *Note:* It is **never** used in electric utility power transmission/distribution systems because at $R_L = R_{\text{Th}}$, the power transfer efficiency is only $\eta = \frac{P_L}{P_{\text{total}}} = \frac{R_L}{R_{\text{Th}}+R_L} = 50\%$ (half of the generated power is wasted as heat inside the source resistance R_{Th}); utility systems instead operate with $R_{\text{Th}} \ll R_L$ to achieve $\eta > 95\%$.



2. Optimizing an Internal Resistor R to Maximize Power in a Fixed Load (2013-14 Q4b, 2024-25 Q7a)

Important Distinction (Sadiku Prob. 4.68, p. 170): When the load resistor $R_{\text{load}} = 10\ \Omega$ is **fixed** and an internal resistor R elsewhere in the circuit is varied (as in 2013-14 Q4b and 2024-25 Q7a, where 12 V is in series with R, in parallel with a middle branch of $10\ \Omega$ and 8 V, and in parallel with an outer $20\ \Omega$ resistor), **do NOT set** $R = R_{\text{Th}}$!

- Instead, the power in the fixed $10\ \Omega$ resistor is $P_{10\Omega} = I_{10\Omega}^2(10\ \Omega) = \frac{(v-8)^2}{10}$, which is maximized when the node voltage v (and current $I_{10\Omega}$) is as large as possible.
- Since $R \geq 0$ is in series with the 12 V source, reducing R to its minimum physical value $\boxed{R = 0\ \Omega}$ (short circuit) directly clamps the top node to $v = 12$ V, giving the maximum current $I_{10\Omega} = \frac{12-8}{10} = 0.4$ A and maximum power $\boxed{P_{\text{max}} = (0.4)^2(10) = 1.6\ \text{W}}$.

3. AC Maximum Average Power Transfer Summary (2016-17 Q4a, 2017-18 Q3b, 2019-20 Q6b)

For a sinusoidal AC circuit with Thevenin impedance $Z_{\text{Th}} = R_{\text{Th}} + jX_{\text{Th}}$ and load impedance $Z_L = R_L + jX_L$:

$$P_{\text{avg}} = \frac{1}{2}|I_{\text{peak}}|^2 R_L = \frac{|V_{\text{Th,peak}}|^2 R_L}{2\left[(R_{\text{Th}} + R_L)^2 + (X_{\text{Th}} + X_L)^2\right]} = \frac{|V_{\text{Th,rms}}|^2 R_L}{(R_{\text{Th}} + R_L)^2 + (X_{\text{Th}} + X_L)^2} \tag{19}$$

1. **Unrestricted Complex Load $Z_L = R_L + jX_L$:** Setting $\frac{\partial P}{\partial X_L} = 0$ and $\frac{\partial P}{\partial R_L} = 0$ gives $X_L = -X_{\text{Th}}$ and $R_L = R_{\text{Th}}$:

$$\boxed{Z_L = Z_{\text{Th}}^* = R_{\text{Th}} - jX_{\text{Th}}} \quad \text{and} \quad \boxed{P_{\text{max}} = \frac{|V_{\text{Th,peak}}|^2}{8R_{\text{Th}}} = \frac{|V_{\text{Th,rms}}|^2}{4R_{\text{Th}}}} \tag{20}$$

2. **Restricted Purely Resistive Load ($X_L = 0$) or Fixed Load with Variable Source Resistor R (2016-17 Q4a):**
   - If $Z_L = R_L$ is purely resistive ($X_L = 0$), setting $\frac{\partial P}{\partial R_L} = 0$ gives $\boxed{R_L = \sqrt{R_{\text{Th}}^2 + X_{\text{Th}}^2} = |Z_{\text{Th}}|}$.
   - If the source resistance R in series with the source is varied while the load is fixed (2016-17 Q4a), average power delivered to the load is maximized when the internal loss resistance is $\boxed{R = 0\ \Omega}$.

4.3 Reciprocity Theorem and Millman's Theorem

Exam Appearances: 2019-20 Q4a (15), 2024-25 Q6c (5), 2024-25 Q7c (5).
Textbook Citations: R. L. Boylestad, *Introductory Circuit Analysis* (12th Ed.), Ch. 9, Sec. 9.6 (Millman's Theorem) & Sec. 9.8 (Reciprocity Theorem); Alexander & Sadiku, Ch. 4, Prob. 4.75 & 4.89, and Ch. 19, Sec. 19.2 (p. 856).

1. State and Explain the Reciprocity Theorem (2024-25 Q6c)

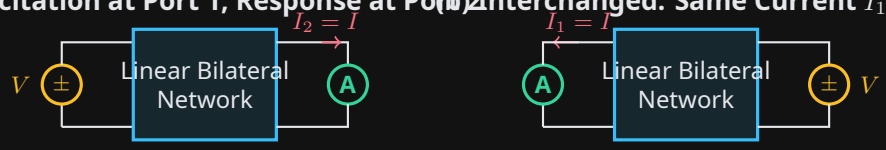
Statement: *"In any linear, bilateral, single-source network (containing no dependent sources), the ratio of the excitation to the response is constant when the positions of the excitation and response are interchanged."*

- **Explanation (Voltage-Source / Ammeter Form):** If a single ideal voltage source V acting in branch 1 (port 1) produces a current reading I in an ideal ammeter (short circuit) located in branch 2 (port 2), then removing the voltage source V from branch 1 (replacing it with a short circuit/ammeter) and placing the exact same voltage source V in branch 2 will produce the **exact same current** I in branch 1.
- **Mathematical Justification:** For a linear passive (bilateral) two-port network, the transfer impedances are equal ($z_{12} = z_{21}$) and the short-circuit transfer admittances are equal ($y_{12} = y_{21}$). Thus:

$$\boxed{\left.\frac{I_2}{V_1}\right|_{V_2=0} = \left.\frac{I_1}{V_2}\right|_{V_1=0} = y_{21} = y_{12}} \quad \text{and dually} \quad \boxed{\left.\frac{V_2}{I_1}\right|_{I_2=0} = \left.\frac{V_1}{I_2}\right|_{I_1=0} = z_{21} = z_{12}} \tag{21}$$

- **Applicability / Limitations:** Applies only to linear, bilateral networks with a **single independent source** (all other independent sources must be zeroed) and **no dependent sources**.

(a) Excitation at Port 1, Response at Port 2 **(b) Interchanged: Same Current $I_1 = I$**



2. State and Explain Millman's Theorem (2024-25 Q7c, 2019-20 Q4a)

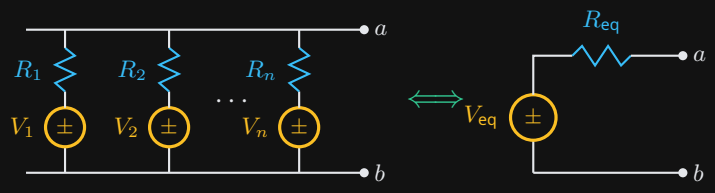
Statement: *"Any number of parallel voltage sources $V_1, V_2, \ldots, V_n$ with series internal resistances $R_1, R_2, \ldots, R_n$ (or impedances $Z_1, Z_2, \ldots, Z_n$) can be replaced by a single equivalent voltage source V_{eq} in series with a single equivalent resistance R_{eq}."*
Mathematical Formula & Proof:

1. Convert each series voltage-source branch (V_k, R_k) into its parallel Norton equivalent current source $I_k = \frac{V_k}{R_k} = G_k V_k$ in parallel with conductance $G_k = \frac{1}{R_k}$.
2. Combine the n parallel current sources and n parallel conductances:

$$I_{\text{eq}} = \sum_{k=1}^{n} I_k = \sum_{k=1}^{n} \frac{V_k}{R_k} = \sum_{k=1}^{n} G_k V_k, \qquad G_{\text{eq}} = \frac{1}{R_{\text{eq}}} = \sum_{k=1}^{n} G_k = \sum_{k=1}^{n} \frac{1}{R_k}$$

3. Convert $(I_{\text{eq}}, R_{\text{eq}})$ back into a single Thevenin voltage source $V_{\text{eq}} = I_{\text{eq}} R_{\text{eq}}$ in series with R_{eq}:

$$\boxed{V_{\text{eq}} = \frac{\displaystyle\sum_{k=1}^{n} \frac{\pm V_k}{R_k}}{\displaystyle\sum_{k=1}^{n} \frac{1}{R_k}} = \frac{G_1 V_1 + G_2 V_2 + \cdots + G_n V_n}{G_1 + G_2 + \cdots + G_n}} \quad \text{and} \quad \boxed{R_{\text{eq}} = \frac{1}{\displaystyle\sum_{k=1}^{n} \frac{1}{R_k}} = \frac{1}{G_1 + G_2 + \cdots + G_n}} \tag{22}$$



Exam Application (2019–20 `Q4a` **— Sadiku Prob. 4.75, p. 171):** Three parallel branches with $V_1 = 1\ \text{V}, V_2 = 2\ \text{V}, V_3 = 3\ \text{V}$ and identical series resistors $R_1 = R_2 = R_3 = R$ feed a load R_L that receives $P_{\text{max}} = 3$ mW. By Millman's theorem:

$$V_{\text{Th}} = V_{\text{eq}} = \frac{\frac{1}{R} + \frac{2}{R} + \frac{3}{R}}{\frac{1}{R} + \frac{1}{R} + \frac{1}{R}} = \frac{6/R}{3/R} = 2\ \text{V}, \qquad R_{\text{Th}} = R_{\text{eq}} = \frac{R}{3}$$

Then $P_{\text{max}} = \frac{V_{\text{Th}}^2}{4R_{\text{Th}}} = \frac{2^2}{4(R/3)} = \frac{3}{R} = 3\ \text{mW} \implies \boxed{R = 1\ \text{k}\Omega}$ (and $R_L = R_{\text{Th}} = 333.3\ \Omega$).

Part III

Energy Storage Elements, Transients, Singularity Functions & Duality (CO2 & CO3)

5 Capacitors, Inductors, Coupled L_{eq} & Singularity Functions

5.1 Capacitors, Inductors & Equivalent Inductance with Dependent Source

Exam Appearances: `2012-13 Q2a`, `2013-14 Q6b (17)`, `2023-24 Q2a (15)`, `2024-25 Q2c (10)`.
Textbook Citations: Alexander & Sadiku, Ch. 6, Sec. 6.2–6.5 (pp. 216–233) & Prob. 6.57 (p. 248).

1. Key Properties of Capacitors and Inductors (Sadiku Table 6.1, p. 232)

- **Capacitor (C):** $q = Cv, \quad i = C\frac{dv}{dt}, \quad v(t) = \frac{1}{C}\int_{t_0}^{t} i(\tau)\,d\tau + v(t_0), \quad w_C = \frac{1}{2}Cv^2 = \frac{q^2}{2C}$.
  - *DC Steady State:* $\frac{dv}{dt} = 0 \implies i = 0$, so a capacitor acts as an **open circuit to DC**.
  - *Continuity:* Capacitor voltage cannot change abruptly: $\boxed{v_C(0^+) = v_C(0^-)}$.
- **Inductor (L):** $v = L\frac{di}{dt}, \quad i(t) = \frac{1}{L}\int_{t_0}^{t} v(\tau)\,d\tau + i(t_0), \quad w_L = \frac{1}{2}Li^2$.
  - *DC Steady State:* $\frac{di}{dt} = 0 \implies v = 0$, so an inductor acts as a **short circuit to DC**.
  - *Continuity:* Inductor current cannot change abruptly: $\boxed{i_L(0^+) = i_L(0^-)}$.

2. Equivalent Inductance L_{eq} of the Coupled/Dependent Network (`2013-14 Q6b`, `2023-24 Q2a`, `2024-25 Q2c`)

Problem (Sadiku Prob. 6.57, p. 248): *Determine L_{eq} at terminals a-b for the network consisting of a 4 H series inductor (carrying current i from a), followed by a parallel combination of a 3 H inductor branch and a second branch containing a dependent voltage source $2\frac{di}{dt}$ ($+$ on left, $-$ on right) in series with a 5 H inductor to ground (b).*
Solution: Let a terminal voltage v_{ab} be applied across terminals a-b, producing input current i through the 4 H inductor. Let i_1 be the current downward through the 3 H inductor, and i_2 be the current through the branch containing the $2\frac{di}{dt}$ source and 5 H inductor:

1. By KCL at the middle node between the 4 H, 3 H, and 5 H inductors:
$$i = i_1 + i_2 \implies \frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt}$$
2. Let $v_1 = 3\frac{di_1}{dt}$ be the voltage across the 3 H inductor. Applying KVL around the right-hand loop gives:
$$v_1 = 3\frac{di_1}{dt} = 2\frac{di}{dt} + 5\frac{di_2}{dt} \implies \frac{di_2}{dt} = \frac{3}{5}\frac{di_1}{dt} - \frac{2}{5}\frac{di}{dt}$$
3. Substituting $\frac{di_2}{dt}$ into the KCL derivative equation $\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt}$:
$$\frac{di}{dt} = \frac{di_1}{dt} + \frac{3}{5}\frac{di_1}{dt} - \frac{2}{5}\frac{di}{dt} \implies \frac{7}{5}\frac{di}{dt} = \frac{8}{5}\frac{di_1}{dt} \implies \frac{di_1}{dt} = \frac{7}{8}\frac{di}{dt}$$
4. Applying KVL from terminal a to terminal b through the 4 H and 3 H inductors:
$$v_{ab} = 4\frac{di}{dt} + 3\frac{di_1}{dt} = 4\frac{di}{dt} + 3\left(\frac{7}{8}\frac{di}{dt}\right) = \left(4 + \frac{21}{8}\right)\frac{di}{dt} = \frac{53}{8}\frac{di}{dt} = L_{\text{eq}}\frac{di}{dt}$$

Hence, the equivalent inductance is $\boxed{L_{\text{eq}} = \frac{53}{8}\ \text{H} = 6.625\ \text{H}}$.

5.2 Singularity Functions: Definitions & Waveform Synthesis

Exam Appearances: `2014-15 Q8a (15)`, `2024-25 Q2a-i,ii (8)`.
Textbook Citations: Alexander & Sadiku, Ch. 7, Sec. 7.4 (pp. 265–272) & Prob. 7.27 (p. 305).

1. Definitions of the Three Singularity Functions

Singularity functions (switching functions) are functions that either are discontinuous or have discontinuous derivatives:

$$\text{1. Unit Step Function:}\quad u(t-t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}, \qquad u(-t) = 1 - u(t) = \begin{cases} 1, & t < 0 \\ 0, & t > 0 \end{cases} \tag{23}$$

$$\text{2. Unit Impulse (Delta) Function:}\quad \delta(t-t_0) = \frac{d}{dt}u(t-t_0) = 0 \quad (t \neq t_0), \qquad \int_{-\infty}^{\infty} f(t)\delta(t-t_0)\,dt = f(t_0) \tag{24}$$

$$\text{3. Unit Ramp Function:}\quad r(t-t_0) = \int_{-\infty}^{t} u(\tau - t_0)\,d\tau = (t-t_0)u(t-t_0) = \begin{cases} 0, & t < t_0 \\ t-t_0, & t \geq t_0 \end{cases} \tag{25}$$

2. Solved Singularity Function Problems (`2014-15 Q8a` and `2024-25 Q2a`)

- **Problem 1 (`2014-15 Q8a`):** *Express $i(t)$ in Fig. for Q. 8(a) in terms of unit step and ramp functions* (where $i(t)$ is 0 for $t < 0$, jumps to $+2$ A for $0 < t < 1$, ramps linearly from $+2$ A at $t = 1$ down to -2 A at $t = 3$ with slope -2 A/s, stays at -2 A for $3 < t < 4$, and returns to 0 for $t > 4$):

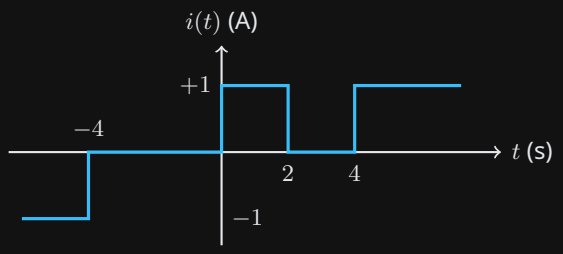
$$\boxed{i(t) = 2u(t) - 2r(t-1) + 2r(t-3) + 2u(t-4)\ \text{A}}$$

- **Problem 2 (`2024-25 Q2a-i` — Sadiku Prob. 7.27, p. 305):** *Express the staircase voltage $v(t)$ in terms of step functions* (where $v(t) = 0$ for $t < -1$, 10 V for $-1 < t < 0$, 30 V for $0 < t < 1$, 20 V for $1 < t < 2$, -10 V for $2 < t < 3$, and 0 V for $t > 3$). Summing the step jumps at each transition point ($+10$ at $t = -1$, $+20$ at $t = 0$, -10 at $t = 1$, -30 at $t = 2$, and $+10$ at $t = 3$):

$$\boxed{v(t) = 10u(t+1) + 20u(t) - 10u(t-1) - 30u(t-2) + 10u(t-3)\ \text{V}}$$

- **Problem 3 (`2024-25 Q2a-ii`):** *Sketch the waveform represented by $i(t) = u(t+4) - u(-t) - u(t-2) + u(t-4)$ A.*
  Piecewise Evaluation: Recall that $u(-t) = 1$ for $t < 0$ and 0 for $t > 0$:

$$i(t) = \begin{cases} 0 - 1 - 0 + 0 = -1\ \text{A}, & t < -4 \\ 1 - 1 - 0 + 0 = 0\ \text{A}, & -4 < t < 0 \\ 1 - 0 - 0 + 0 = +1\ \text{A}, & 0 < t < 2 \\ 1 - 0 - 1 + 0 = 0\ \text{A}, & 2 < t < 4 \\ 1 - 0 - 1 + 1 = +1\ \text{A}, & t > 4 \end{cases}$$


$i(t)$ (A)
$+1$
-4
t (s)
2
4
-1

6 First-Order & Second-Order Transient Circuits and Duality

6.1 First-Order Step Response Derivation & Energy Time-Constant Proof

Exam Appearances: `2016-17 Q2c (8)`, `2022-23 Q4b-iii`.
Textbook Citations: Alexander & Sadiku, Ch. 7, Sec. 7.3 & Sec. 7.6 (pp. 260, 280–281).

1. Derivation of Inductor Current $i(t)$ in Response to a Step Voltage in a Series RL Circuit (`2016-17 Q2c`)

Consider a series RL circuit driven by a step voltage source $v_s(t) = V_s u(t)$ (or $1 \cdot u(t)$ in Fig. Q. 2c) with initial inductor current $i(0^-) = i(0^+) = I_0$ (where $I_0 = 0$ if initially uncharged).

1. Applying KVL around the loop for $t > 0$ (where $u(t) = 1$):

$$L\frac{di}{dt} + Ri = V_s \implies \frac{di}{dt} + \frac{R}{L}i = \frac{V_s}{L}$$

2. Separating variables:

$$\frac{di}{i - \frac{V_s}{R}} = -\frac{R}{L}\,dt$$

3. Integrating both sides from $t = 0$ (where $i(0) = I_0$) to t:

$$\int_{I_0}^{i(t)} \frac{di'}{i' - \frac{V_s}{R}} = -\frac{R}{L}\int_0^t dt' \implies \ln\left(\frac{i(t) - \frac{V_s}{R}}{I_0 - \frac{V_s}{R}}\right) = -\frac{R}{L}t = -\frac{t}{\tau} \qquad \left(\text{where } \tau = \frac{L}{R}\right)$$

4. Exponentiating both sides yields the complete step response:

$$\boxed{i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-t/\tau}, \quad t > 0} \qquad \left(\text{and for } I_0 = 0 \text{ and } V_s = 1\text{ V: } \boxed{i(t) = \frac{1}{R}\left(1 - e^{-(R/L)t}\right)u(t)}\right) \tag{26}$$

2. How Many Time Constants Does It Take to Deliver 95% of the Total Energy? (`2022-23 Q4b-iii`)

In a source-free RL or RC circuit, current and voltage decay as $i(t) = I_0 e^{-t/\tau}$, so the instantaneous power delivered to a resistor R is proportional to $e^{-2t/\tau}$:

$$p(t) = i^2(t)R = I_0^2 R e^{-2t/\tau}$$

The energy $W(t)$ delivered to the resistor from 0 to t as a fraction of the total energy $W(\infty) = \int_0^\infty I_0^2 R e^{-2t'/\tau}dt' = \frac{1}{2}\tau I_0^2 R$ is:

$$\frac{W(t)}{W(\infty)} = \frac{\int_0^t I_0^2 R e^{-2t'/\tau}dt'}{\frac{1}{2}\tau I_0^2 R} = 1 - e^{-2t/\tau}$$

Setting $\frac{W(t)}{W(\infty)} = 0.95$ (95%):

$$1 - e^{-2t/\tau} = 0.95 \implies e^{-2t/\tau} = 0.05 \implies \frac{t}{\tau} = -\frac{\ln(0.05)}{2} = \frac{\ln 20}{2} \approx \boxed{1.498 \approx 1.5 \text{ time constants}} \tag{27}$$

6.2 Second-Order RLC Circuits, Airbag Igniter Peak Time & Circuit Duality

Exam Appearances: `2020-21 Q5b`, `2022-23 Q1a`, `2024-25 Q3c (5)`, `2024-25 Q4b (18)`.
Textbook Citations: Alexander & Sadiku, Ch. 8, Sec. 8.2–8.4, Sec. 8.10 (Duality, pp. 350–352), & Prob. 8.78 (p. 367).

1. Summary of Second-Order RLC Damping Regimes

- **Series RLC Circuit:** Characteristic equation $s^2 + \frac{R}{L}s + \frac{1}{LC} = s^2 + 2\alpha s + \omega_0^2 = 0$, where $\boxed{\alpha = \frac{R}{2L}}$ and $\boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$.
- **Parallel RLC Circuit:** Characteristic equation $s^2 + \frac{1}{RC}s + \frac{1}{LC} = s^2 + 2\alpha s + \omega_0^2 = 0$, where $\boxed{\alpha = \frac{1}{2RC}}$ and $\boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$.
- **Three Damping Cases** with roots $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$ and damped frequency $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$:
  1. *Overdamped ($\alpha > \omega_0$):* $x(t) = x(\infty) + A_1 e^{s_1 t} + A_2 e^{s_2 t}$.
  2. *Critically Damped ($\alpha = \omega_0$):* $x(t) = x(\infty) + (A_1 + A_2 t)e^{-\alpha t}$.
  3. *Underdamped ($\alpha < \omega_0$):* $x(t) = x(\infty) + e^{-\alpha t}\,(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$.

2. Automobile Airbag Igniter First-Peak Time (`2024-25 Q4b` — Sadiku Prob. 8.78, p. 367)

Problem: *In Fig. 4(b), the capacitor $C = \frac{1}{30}$ F is charged to 12 V in position A. At $t = 0$, the switch moves to B, placing C, $L = 60$ mH $= 0.06$ H, and the airbag igniter resistor $R = 3\,\Omega$ in a source-free series RLC loop. Find the time it takes the voltage $v_R(t)$ across the igniter R to reach its first peak.*
Solution:

1. **Initial Conditions at $t = 0^+$:** $v_C(0^+) = v_C(0^-) = 12$ V and $i(0^+) = i(0^-) = 0$ A. Therefore, $v_R(0^+) = Ri(0^+) = 0$ V, and since $v_L(0^+) = v_C(0^+) - v_R(0^+) = 12$ V, we have $\frac{di(0^+)}{dt} = \frac{v_L(0^+)}{L} = \frac{12}{0.06} = 200$ A/s.
2. **Damping Parameters:** For this series RLC circuit:

   $$\alpha = \frac{R}{2L} = \frac{3}{2(0.06)} = 25 \text{ Np/s}, \qquad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.06 \times \frac{1}{30}}} = \frac{1}{\sqrt{0.002}} = \sqrt{500} \approx 22.361 \text{ rad/s}$$

   Since $\alpha = 25 > \omega_0 = 22.361$, the circuit is **overdamped** with roots:

   $$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -25 \pm \sqrt{625 - 500} = -25 \pm \sqrt{125} = -25 \pm 5\sqrt{5} \implies s_1 \approx -13.8197, \quad s_2 \approx -36.1803$$

3. **Current and Igniter Voltage:** Since $i(0^+) = A_1 + A_2 = 0 \implies A_2 = -A_1$, we have $i(t) = A_1(e^{s_1 t} - e^{s_2 t})$ and $v_R(t) = Ri(t) = 3A_1(e^{s_1 t} - e^{s_2 t})$.
4. **Time of First Peak ($\frac{dv_R}{dt} = 0$):** Differentiating $v_R(t)$ and setting to zero:

   $$s_1 e^{s_1 t_p} - s_2 e^{s_2 t_p} = 0 \implies e^{(s_1 - s_2)t_p} = \frac{s_2}{s_1} \implies \boxed{t_p = \frac{\ln(s_2/s_1)}{s_1 - s_2} = \frac{\ln\left(\frac{36.1803}{13.8197}\right)}{22.3607} \approx 0.0430 \text{ s} = 43.0 \text{ ms}}$$

3. Principle of Duality and Solution to `2024-25 Q3c` (Sadiku Sec. 8.10, pp. 350–352)

Two planar circuits are **duals** if the mesh equations of one have the exact same mathematical form as the node equations of the other:

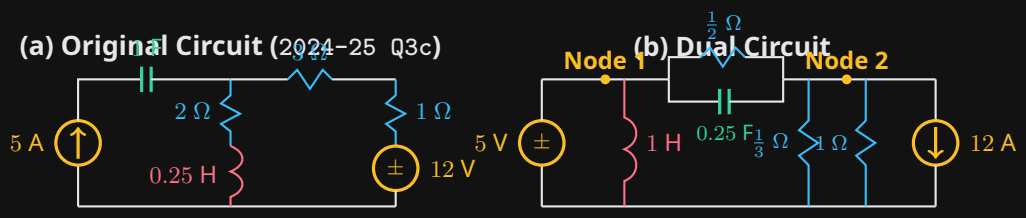
Quantity	Dual Quantity	Quantity	Dual Quantity
Resistance R (Ω)	Conductance G (S) or $R' = 1/R$	Mesh / Loop	Non-reference Node
Inductance L (H)	Capacitance C (F)	Series Connection	Parallel Connection
Capacitance C (F)	Inductance L (H)	Open Circuit	Short Circuit
Voltage Source v_s (V)	Current Source i_s (A)	KVL / Thevenin	KCL / Norton

Solution to `2024-25 Q3c`**:** The given circuit has two meshes:

- **Mesh 1 (left)** has a 5 A upward current source on the left, a 1 F capacitor on the top, and shares a $2\ \Omega$ resistor in series with a 0.25 H inductor with Mesh 2.
- **Mesh 2 (right)** shares the $2\ \Omega + 0.25$ H central branch with Mesh 1, has a $3\ \Omega$ resistor on the top, and has a 12 V voltage source ($+$ on top) in series with a $1\ \Omega$ resistor on the right.

Placing Node 1 in Mesh 1, Node 2 in Mesh 2, and Reference Node 0 outside gives the **Dual Circuit**:

- **Between Node 1 and Ground (0):** A 5 V voltage source ($+$ at Node 1) in parallel with a 1 H inductor (dual of the 5 A source and 1 F capacitor).
- **Between Node 1 and Node 2:** A $\frac{1}{2}\ \Omega$ resistor ($G = 2$ S) in parallel with a 0.25 F capacitor (dual of the series $2\ \Omega$ resistor and 0.25 H inductor).
- **Between Node 2 and Ground (0):** A $\frac{1}{3}\ \Omega$ resistor ($G = 3$ S) in parallel with the parallel combination of a $1\ \Omega$ resistor ($G = 1$ S) and a 12 A current source pointing downward from Node 2 to Ground 0 (since the 12 V source opposes clockwise mesh current i_2).



Part IV

Magnetic Circuits & AC Steady-State / Power Analysis (CO3 & CO4)

7 Magnetic Circuits: Definitions, Laws, Analogy & Working Formulas

7.1 Flux Density, Permeability, Magnetizing Force & Ampere's Circuital Law

Exam Appearances: `2010-11 Q8a`, `2011-12 Q4a,b,c (35)`, `2013-14 Q8a,b`, `2014-15 Q8b`, `2020-21 Q8a,b`, `2021-22 Q4a,b`, `2022-23 Q1b, Q4a`, `2023-24 Q1b`, `2024-25 Q1b, Q2b`.
Textbook Citations: R. L. Boylestad, *Introductory Circuit Analysis* (12th Ed.), Ch. 11 (Magnetic Circuits).

1. Define Magnetic Flux Density, Permeability, Magnetizing Force & Reluctance (`2011-12 Q4a`, `2024-25 Q1b`)

1. **Magnetic Flux (Φ) and Magnetic Flux Density (B):**
   - *Magnetic Flux (Φ):* The total number of magnetic field lines passing through a given surface area, measured in **webers (Wb)**.
   - *Magnetic Flux Density (B):* The magnetic flux passing perpendicularly through a unit cross-sectional area A of the magnetic core, measured in **teslas (T)** or **webers per square meter (Wb/m^2)**:

$$\boxed{B = \frac{\Phi}{A}} \qquad \left(\text{Unit: } 1\ \text{T} = 1\ \text{Wb/m}^2\right) \tag{28}$$

2. **Permeability (μ, μ_0, μ_r):**
   - *Permeability (μ):* A measure of the ease with which magnetic flux lines can be established in a material, defined as the ratio of magnetic flux density B to magnetizing force H, measured in **henrys per meter (H/m)** or $\text{Wb}/(\text{A}\cdot\text{m})$:

$$\boxed{\mu = \frac{B}{H} = \mu_0\mu_r} \qquad \text{where } \mu_0 = 4\pi \times 10^{-7}\ \text{H/m (permeability of free space/air)} \tag{29}$$

   - *Relative Permeability ($\mu_r = \frac{\mu}{\mu_0}$):* The dimensionless ratio of the permeability of a material to the permeability of free space ($\mu_r = 1$ for air/vacuum; $\mu_r \gg 1$ for ferromagnetic materials like sheet steel, cast steel, and cast iron, where $\mu = B/H$ varies nonlinearly along the B-H magnetization curve).

3. **Magnetizing Force / Magnetic Field Intensity (H):** The magnetomotive force ($\mathcal{F} = NI$) per unit length l of the magnetic circuit required to establish a given flux density B, measured in **ampere-turns per meter (At/m or A/m)**:

$$\boxed{H = \frac{\mathcal{F}}{l} = \frac{NI}{l} = \frac{B}{\mu}} \qquad \left(\text{In an air gap: } H_g = \frac{B_g}{\mu_0} = \frac{B_g}{4\pi \times 10^{-7}}\ \text{At/m}\right) \tag{30}$$

4. **Magnetomotive Force ($\mathcal{F}$), Reluctance ($\mathcal{R}$), and Ohm's Law for Magnetic Circuits:**

$$\boxed{\mathcal{F} = NI = \Phi\mathcal{R} = Hl}, \qquad \text{where } \boxed{\mathcal{R} = \frac{l}{\mu A}\quad (\text{At/Wb})} \quad \text{and} \quad \boxed{\Phi = \frac{\mathcal{F}}{\mathcal{R}} = \frac{NI}{\mathcal{R}}} \tag{31}$$

2. Explain Ampere's Circuital Law for Magnetic Circuits (2011-12 Q4b)

Statement: *"The algebraic sum of the rises and drops of magnetomotive force ($\mathcal{F} = Hl = \Phi\mathcal{R}$) around any closed loop of a magnetic circuit is equal to zero; equivalently, the line integral of the magnetizing force $\vec{H}$ around any closed magnetic path equals the net algebraic ampere-turns (NI) enclosed by that path."*

$$\boxed{\oint \vec{H} \cdot d\vec{l} = \sum_{k=1}^{m} H_k l_k = \sum NI \quad \Longleftrightarrow \quad \sum \mathcal{F}_{\text{sources}} - \sum H_k l_k = 0} \tag{32}$$

Explanation: Ampere's circuital law in magnetic circuits is the direct counterpart of **Kirchhoff's Voltage Law (KVL)** in electric circuits:

- For a **series magnetic circuit** with sections of lengths $l_1, l_2, \ldots$ and an air gap of length l_g (carrying common flux Φ), Ampere's circuital law gives:

  $$NI = H_1 l_1 + H_2 l_2 + H_g l_g \qquad (\text{or } N_1 I_1 \pm N_2 I_2 = H_{\text{steel}} l_{\text{steel}} + H_{\text{iron}} l_{\text{iron}} \text{ as in 2024-25 Q1b})$$

- For a **series-parallel magnetic circuit** (such as a two-window core in 2010-11 Q8a, 2011-12 Q4c, 2022-23 Q1b, 2023-24 Q1b), the flux junction obeys Gauss's law for magnetism (**KCL analogue**: $\boxed{\Phi_T = \Phi_1 + \Phi_2}$), and Ampere's circuital law is applied around each closed magnetic window loop.

Electric Circuit Quantity	Magnetic Circuit Analogue	Defining Relation
Electromotive Force / Voltage (V or E, V)	Magnetomotive Force ($\mathcal{F} = NI$, At)	Drives flux / current
Electric Current (I, A)	Magnetic Flux (Φ, Wb)	$\Phi = BA \leftrightarrow I = JA$
Current Density ($J = I/A$, A/m^2)	Magnetic Flux Density ($B = \Phi/A$, T)	Flux per unit area
Electric Field Intensity ($E = V/l$, V/m)	Magnetizing Force ($H = NI/l$, At/m)	MMF drop per unit length
Electrical Conductivity (σ, S/m)	Permeability ($\mu = \mu_0\mu_r$, H/m)	$B = \mu H \leftrightarrow J = \sigma E$
Electrical Resistance ($R = \frac{l}{\sigma A}$, Ω)	Magnetic Reluctance ($\mathcal{R} = \frac{l}{\mu A}$, At/Wb)	$\Phi = \frac{\mathcal{F}}{\mathcal{R}} \leftrightarrow I = \frac{V}{R}$
Kirchhoff's Laws ($\sum I = 0$, $\sum V = 0$)	$\sum \Phi = 0$, Ampere's Law ($\sum NI = \sum H_k l_k$)	Node & Loop Laws

Note on Unit Conversions in Magnetic Circuit Problems: 1 inch $= 2.54$ cm $= 0.0254$ m and 1 inch^2 $= 6.4516 \times 10^{-4}$ m^2.

7.2 AC Steady-State Phasor Diagram, RMS Values & Power Factor Correction

Exam Appearances: 2015-16 Q7a,b, 2016-17 Q2b (12), Q4b (20), 2017-18 Q4a,b (35), 2018-19 Q7b, Q8a, 2019-20 Q7b, Q8a,b.
Textbook Citations: Alexander & Sadiku, Ch. 9, 10, & Ch. 11, Sec. 11.4–11.8 (pp. 467–483).

1. Average Value, RMS (Effective) Value & Form Factor of Periodic Waveforms

For any periodic waveform $v(t)$ with period T:

$$\boxed{V_{\text{avg}} = \frac{1}{T}\int_0^T v(t)\,dt}, \qquad \boxed{V_{\text{rms}} = V_{\text{eff}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}}, \qquad \boxed{P_{\text{avg}} = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R}} \tag{33}$$

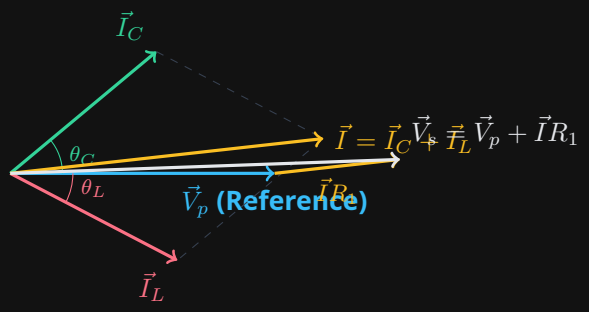
(Useful Shortcut for Piecewise Segments over Interval Δt: A constant segment V_0 contributes $V_0^2\Delta t$ to $\int v^2 dt$; a linear triangle/ramp from 0 to V_m over Δt contributes $\frac{1}{3}V_m^2\Delta t$; and a half-sine arch $V_m \sin\omega t$ over Δt contributes $\frac{1}{2}V_m^2\Delta t$).*

2. Qualitative Phasor Diagram for Series-Parallel RLC Circuit (`2016-17 Q2b`)

Problem: *Qualitatively draw the phasor diagram for I, I_C, I_L, V_p and V_s in Fig. Q. 2b, where source V_s is in series with R_1 (carrying total current I) and feeds a parallel tank (voltage V_p) consisting of a capacitive branch (C_1 in series with R_C, carrying I_C) and an inductive branch (L_1 in series with R_L, carrying I_L).*

Step-by-Step Phasor Construction:

1. Choose the parallel branch voltage $\vec{V}_p = V_p\angle 0^\circ$ as the horizontal reference phasor.
2. In the capacitive branch ($Z_C = R_C - jX_{C1}$), the current $\vec{I}_C = \frac{\vec{V}_p}{R_C - jX_{C1}}$ **leads** $\vec{V}_p$ by an acute angle $\theta_C = \tan^{-1}\left(\frac{X_{C1}}{R_C}\right) < 90^\circ$.
3. In the inductive branch ($Z_L = R_L + jX_{L1}$), the current $\vec{I}_L = \frac{\vec{V}_p}{R_L + jX_{L1}}$ **lags** $\vec{V}_p$ by an acute angle $\theta_L = \tan^{-1}\left(\frac{X_{L1}}{R_L}\right) < 90^\circ$.
4. By KCL, the total source current is the vector (phasor) sum $\vec{I} = \vec{I}_C + \vec{I}_L$.
5. By KVL, the input source voltage is the vector sum $\vec{V}_s = \vec{V}_p + \vec{I}R_1$, where $\vec{I}R_1$ is parallel to (in phase with) $\vec{I}$.



3. Complex Power & Power Factor Correction Formulas (`2016-17 Q4b, 2017-18 Q4b, 2018-19 Q7b, 2019-20 Q8a,b`)

For a system of parallel AC loads supplied at line voltage V_{rms} and frequency f ($\omega = 2\pi f$):

$$S_{\text{total}} = \sum P_k + j\sum Q_k = P_T + jQ_T = |S_T|\angle\theta_1 = V_{\text{rms}} I_{\text{rms}}^*, \qquad \text{pf}_{\text{old}} = \cos\theta_1 = \frac{P_T}{\sqrt{P_T^2 + Q_T^2}}$$

$$Q_k = P_k \tan(\cos^{-1}\text{pf}_k) \quad (+Q_k \text{ for lagging/inductive pf}, \ -Q_k \text{ for leading/capacitive pf})$$

To improve the overall lagging power factor from $\cos\theta_1$ to a higher lagging power factor $\cos\theta_2$ (without altering the total real power P_T), connect a shunt capacitor C in parallel across the load terminals providing reactive power Q_C:

$$\boxed{Q_C = Q_1 - Q_2 = P_T(\tan\theta_1 - \tan\theta_2) = \frac{V_{\text{rms}}^2}{X_C} = \omega C V_{\text{rms}}^2} \implies \boxed{C = \frac{P_T(\tan\theta_1 - \tan\theta_2)}{2\pi f V_{\text{rms}}^2}} \qquad (34)$$

*(Note: For correction to **unity power factor**, $\cos\theta_2 = 1 \implies \tan\theta_2 = 0$, so $Q_C = Q_1 = P_T\tan\theta_1$ and $C = \frac{Q_1}{2\pi f V_{\text{rms}}^2}$).*

References & Textbook Citations

1. **Alexander, Charles K. & Sadiku, Matthew N. O.**, *Fundamentals of Electric Circuits* (5th Edition), McGraw-Hill Education (Primary BUET EEE 101 Textbook).
2. **Nilsson, James W. & Riedel, Susan A.**, *Electric Circuits* (10th Edition), Pearson Education.
3. **Dorf, Richard C. & Svoboda, James A.**, *Introduction to Electric Circuits* (9th Edition), John Wiley & Sons.
4. **Boylestad, Robert L.**, *Introductory Circuit Analysis* (12th/13th Edition), Pearson Education (Primary Reference for Magnetic Circuits, Static/Dynamic Resistance, Reciprocity Theorem & Millman's Theorem).